ScAN - A Path to PIM Hardware-aware Network

Phase Roadmap

Phase 1 Float Training AllCNN-C CIFAR-100 FP32	**Phase 2** W4A4 QAT uniform W4 PACT activations integer simulator	**Phase 3** KD Finetune EfficientNet teacher SS / CS / TU	**Phase 4** PIM-QAT dense N=144 tile ADC7, no noise paper setting
Phase 5 PCN Integration 3-iter fb/ff refine sensitive channels	**Phase 6** HW Noise Finetune real F/G/H LUT voltage readout	**Final Eval** HW simulator full noise model multi-seed	

Phase Results

Phase	Experiment	Precision / path	Acc. (%)	Notes
1	RGB AllCNN-C + BN	FP32 float	72.99	Standard CIFAR-100 RGB, 3x32x32, crop/flip augmentation.
1	RAW/RGGB AllCNN-C	FP32 float	61.31	Current 4-channel RGGB BN baseline, raw_noise=True.
2	RAW/RGGB W4A4 SS	uniform W4A4 INT sim	56.61	Main noisy self-studying baseline; use uniform path going forward.
2	RAW/RGGB W4A4 SS + PACT	uniform I16/W4/A4 INT sim	60.01	qkd_use_lutq=False; gain comes from PACT/I16 and simulator alignment.
3	QKD CS + PACT	I16/W4/A4 equal-spaced LUT-Q m=15	61.24	Previous co-studying simulator result.
3	QKD TU + PACT	I16/W4/A4 equal-spaced LUT-Q m=15	61.66	Current best dense int4 simulator result.
4	PIM-QAT dense N144	paper W4/A4/ADC7 simulator	56.75	Self-study from FP32, uniform W4 + PACT, N=144, ADC7, no noise; GPU forward best 56.88.

Teacher / KD

Model / stage	Acc. (%)	Notes
EfficientNetV2-L teacher, clean	77.30	Verified on current H5 test split.
EfficientNetV2-L teacher, noisy	77.33	Default noisy evaluation.
Co-studied teacher, noisy	78.70	Earlier CS run improved teacher.
Uniform SS + PACT rerun	60.01	I16/W4/A4, qkd_use_lutq=False, fixed integer simulator.
QKD CS rerun	61.24	I16/W4/A4 equal-spaced LUT-Q + PACT, fixed simulator.
QKD TU rerun	61.66	Best current dense int4 simulator result.

Current PIM Setting

Item	Current setting
Model	ALL-CNN-C, 4-channel RGGB input, BatchNorm; GBN and PCN disabled for the minimal pass.
Weights	Uniform 4-bit deployed integer code range [-7,...,7] with precomputed per-output-channel scales.
Activations	First layer input is unsigned I16; hidden activations are unsigned A4 codes in [0,15] and values in [0,1].
Tile	Hidden PIM layers use dense N=144 tiles. Stage1.0 and logits stay digital for this pass.
ADC	7-bit paper PIM quantizer for bit-serial partial sums, forward scale 1.03, no read noise.
Simulator	Full test uses the PIM simulator with integer weights, integer activations, precomputed scales, and no runtime tanh.

Next

• Improve Phase 4 dense PIM-QAT beyond the current 56.75% simulator result.

• Add pim_info/harware_model.tex voltage-domain readout once F/G/H and noise parameters are fixed.

• Add PCN only after the minimal BN PIM path is stable.